

Project Report

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1 Project Report

2 Grocery Inventory & Sales Performance Dashboard

2.1 Microsoft Excel Data Analytics Project

Project Name: Grocery Inventory & Sales Performance Dashboard

Domain: Grocery Retail & Inventory Management

Tool Used: Microsoft Excel

Project Type: Data Analytics Portfolio Project

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4 1. Executive Summary

The Grocery Inventory & Sales Performance Dashboard is an end-to-end Microsoft Excel Data Analytics project developed to transform raw grocery inventory and sales data into meaningful business insights.

The project follows the complete analytics lifecycle, including:

- Business requirement analysis
- Data cleaning
- Data transformation
- KPI development
- Pivot Table creation
- Pivot Chart development
- Interactive dashboard design
- Business insight generation

The final dashboard enables management to monitor inventory performance, analyze revenue trends, identify products requiring replenishment, and evaluate category-level business performance through interactive visualizations.

5 2. Project Background

Retail grocery businesses manage hundreds of products across multiple categories while maintaining the right balance between inventory availability and customer demand.

Without an effective reporting system, managers often struggle to:

- Monitor sales performance.
- Track inventory value.
- Identify stock shortages.
- Compare category performance.
- Monitor monthly revenue.
- Make timely replenishment decisions.

To address these challenges, an interactive dashboard was developed using Microsoft Excel.

6 3. Business Problem

The organization maintained inventory and sales records in Excel but lacked a centralized reporting system.

Business users relied on manual reports, making it difficult to analyze operational performance.

Major challenges included:

- No centralized dashboard.
- Manual report preparation.
- Difficulty identifying high-performing categories.

- Lack of monthly trend analysis.
 - Poor visibility into inventory investment.
 - Difficulty identifying products requiring replenishment.
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7 4. Project Objectives

The project was developed with the following objectives:

- Build an interactive Excel dashboard.
 - Monitor inventory performance.
 - Track monthly revenue trends.
 - Compare category performance.
 - Analyze inventory value.
 - Identify products requiring replenishment.
 - Monitor inventory turnover.
 - Support business decision-making.
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8 5. Dataset Overview

The dashboard was developed using a grocery inventory dataset.

8.0.1 Dataset Summary

Item	Details
Domain	Grocery Retail
Records	990 Products
Categories	7
Analysis Period	2024–2025
Tool	Microsoft Excel

8.0.2 Main Dataset Fields

- Product_ID
- Product_Name
- Category
- Supplier_ID
- Supplier_Name
- Stock_Quantity
- Reorder_Level
- Reorder_Quantity
- Unit_Price
- Sales_Volume
- Inventory_Turnover_Rate
- Date_Received

- Last_Order_Date
 - Expiration_Date
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9 6. Data Cleaning & Transformation

Before dashboard development, the dataset underwent several cleaning and transformation activities.

9.0.1 Data Cleaning Activities

- Standardized mixed date formats.
- Converted Unit Price from text to numeric.
- Created Revenue column.
- Created Inventory Value column.
- Created Reorder Status column.
- Created Month_Year helper column.
- Validated data types.
- Refreshed Pivot Tables.

9.0.2 Calculated Columns

- Revenue
- Inventory_Value
- Reorder_Status
- Month_Year

These transformations ensured accurate calculations and reliable dashboard reporting.

10 7. Dashboard Development

The dashboard was developed using Microsoft Excel.

10.0.1 Excel Features Used

- Excel Tables
- Pivot Tables
- Pivot Charts
- Slicers
- Timeline Filter
- Conditional Formatting
- Excel Formulas
- Shapes & Icons
- Custom Number Formatting

Interactive filtering was implemented to improve dashboard usability.

11 8. Dashboard Features

The dashboard provides an interactive overview of business performance.

11.0.1 KPI Cards

- Total Products
- Total Revenue
- Total Units Sold
- Inventory Value
- Reorder Products
- Reorder Percentage
- Average Inventory Turnover Rate

11.0.2 Interactive Filters

- Category
- Year Timeline
- Reorder Status

These filters dynamically update all visualizations.

12 9. Key Performance Indicators (KPIs)

KPI	Business Purpose
Total Products	Inventory size
Total Revenue	Overall sales performance
Total Units Sold	Customer demand
Inventory Value	Current inventory investment
Reorder Products	Products requiring replenishment
Reorder Percentage	Inventory health indicator
Average Inventory Turnover Rate	Inventory efficiency

These KPIs provide a quick summary of business performance.

13 10. Dashboard Visualizations

The dashboard contains six major visualizations.

13.1 Monthly Revenue Trend

Displays monthly revenue performance across the selected year.

Business Value:

- Monitor monthly growth.
 - Identify seasonal demand.
 - Compare revenue over time.
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13.2 Revenue by Category

Compares revenue generated by each product category.

Business Value:

- Identify high-performing categories.
 - Support promotional planning.
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13.3 Inventory Value by Category

Displays inventory investment across product categories.

Business Value:

- Monitor inventory allocation.
 - Improve stock planning.
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13.4 Top 10 Products by Revenue

Ranks products based on revenue generated.

Business Value:

- Identify best-selling products.
 - Support marketing decisions.
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13.5 Inventory Turnover Rate by Category

Compares inventory movement across categories.

Business Value:

- Evaluate inventory efficiency.
 - Identify slow-moving categories.
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13.6 Reorder Status Distribution

Displays the proportion of products requiring replenishment.

Business Value:

- Monitor stock health.

- Improve procurement planning.
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14 11. Business Insights

The dashboard provides valuable insights for management.

14.0.1 Sales Insights

- Total business revenue.
- Monthly revenue trends.
- Top-performing categories.
- Highest revenue-generating products.

14.0.2 Inventory Insights

- Current inventory value.
- Inventory turnover performance.
- Products requiring replenishment.
- Reorder percentage.

14.0.3 Operational Insights

- Category-level performance.
 - Interactive year-wise comparison.
 - Business performance through dynamic filters.
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15 12. Challenges Faced

Several practical challenges were encountered during the project.

15.0.1 Challenge 1

Mixed date formats.

Solution:

Standardized all date fields using Excel functions.

15.0.2 Challenge 2

Currency stored as text.

Solution:

Converted Unit Price into numeric values.

15.0.3 Challenge 3

Missing business metrics.

Solution:

Created Revenue and Inventory Value columns.

15.0.4 Challenge 4

Monthly trend analysis.

Solution:

Created the Month_Year helper column and implemented a Timeline filter.

15.0.5 Challenge 5

Pivot Table refresh issues after adding calculated columns.

Solution:

Updated the data source and refreshed all Pivot Tables before reconnecting slicers.

16 13. Business Recommendations

Based on dashboard insights, the following recommendations are suggested.

- Monitor Monthly Revenue Trend regularly.
 - Prioritize replenishment for products below the reorder level.
 - Increase focus on high-performing categories.
 - Optimize inventory investment for slow-moving categories.
 - Review inventory turnover periodically.
 - Use dashboard filters for detailed business analysis.
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17 14. Learning Outcomes

This project enhanced practical knowledge in:

17.0.1 Technical Skills

- Microsoft Excel
- Data Cleaning
- Data Transformation
- Excel Formulas
- Pivot Tables

- Pivot Charts
- Dashboard Design
- KPI Development

17.0.2 Analytical Skills

- Business Analysis
- Inventory Analysis
- Sales Analysis
- Data Visualization
- Report Development

17.0.3 Professional Skills

- Problem Solving
 - Analytical Thinking
 - Documentation
 - Business Communication
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18 15. Future Enhancements

Future versions of the dashboard may include:

- Profit & Margin Analysis
 - Supplier Performance Dashboard
 - Expiry Risk Monitoring
 - Forecasting
 - SQL Database Integration
 - Power BI Dashboard
 - Python-based Predictive Analytics
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19 16. Conclusion

The Grocery Inventory & Sales Performance Dashboard successfully demonstrates the complete data analytics workflow using Microsoft Excel.

Beginning with raw inventory data, the project progressed through data cleaning, transformation, KPI development, Pivot Table analysis, interactive dashboard creation, and business insight generation.

The final dashboard enables decision-makers to monitor inventory health, analyze monthly revenue trends, evaluate product performance, and support inventory planning through an intuitive reporting interface.

This project demonstrates strong practical skills in **Excel, data cleaning, business analysis, dashboard development, KPI design, and data visualization**, making it a comprehensive portfolio project for aspiring Data Analysts.